THUE'S EQUATION, WITH SOLUTIONS

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This worksheet works through [Sil09, Exercises 9.5 and 9.6(a)–(b)]. In 1909 Thue proved that an equation $F(x, y) = b$, where F is a binary form of degree at least 3 with integer coefficients and nonzero discriminant and b is a nonzero integer, has only finitely many integer solutions. His idea was to turn the equation into a problem of Diophantine approximation. A solution with y large makes x/y very close to a root of $F(T, 1)$, and algebraic numbers cannot be approximated too well by rational numbers. Problem 1.6 makes the first step quantitative: a polynomial is small only near its roots. Problem 2.5 proves Thue's theorem with Roth's theorem, and extends it to the S-integers of a number field for forms with an irreducible factor of degree at least 3. One part of each exercise is corrected or restricted, as explained at the start of each section.

Standing notation.

- A number field K, its set of places $\mathcal{P}_K$, and the subset $\mathcal{P}_K^\infty$ of archimedean places.
- For $v \in \mathcal{P}_K$, the completion K_v, the absolute value $|\ |_v$ on K_v extending the usual absolute value on $\mathbb{Q}_p$ or $\mathbb{R}$, and the local degree $n_v \coloneqq [K_v : \mathbb{Q}_v]$, where $\mathbb{Q}_v$ is the completion of $\mathbb{Q}$ at the restriction of v.
- An algebraic closure $\overline{K}$ of K. For $v \in \mathcal{P}_K$, an algebraic closure $\overline{K_v}$ of K_v, to which $|\ |_v$ extends uniquely, as K_v is complete, and a K-embedding $\iota_v \colon \overline{K} \to \overline{K_v}$. Then $|z|_v \coloneqq |\iota_v(z)|_v$ extends $|\ |_v$ to $\overline{K}$. The ι_v are arbitrary, and every statement below holds for every choice of them. For $K = \mathbb{Q}$, the field $\overline{\mathbb{Q}}$ is the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$, ι_∞ is the inclusion $\overline{\mathbb{Q}} \subseteq \mathbb{C}$, and $|\ |$ is the usual absolute value.
- The height $H_K(P) \coloneqq \prod_{v \in \mathcal{P}_K} \max_i |x_i|_v^{n_v}$ of a point $P = [x_0 : \cdots : x_N]$ of $\mathbb{P}^N(K)$, which does not depend on the coordinates and satisfies $H_K(P) \geqslant 1$ [Sil09, §VIII.5]. For $x \in K$, $H_K(x) \coloneqq H_K([x : 1])$. For coprime integers p and q with $q \neq 0$, $H_{\mathbb{Q}}(p/q) = \max(|p|, |q|)$.
- The product formula $\prod_{v \in \mathcal{P}_K} |a|_v^{n_v} = 1$ for $a \in K^\times$ [Sil09, Product Formula VIII.5.3].

How to use this. Allow about 30 minutes for Problem 1.6 and 35 for Problem 2.5. Problem 2.5 uses only Problem 1.6(1) and (3), so Problem 1.6(4) and (5) can wait for a second pass. The `Magma` block runs in seconds in the free online calculator. Everything else is by hand.

1. Polynomials are small only near their roots

Problem 1.6 is [Sil09, Exercise 9.5]. If t is far from every root of a polynomial f, then $|f(t)|$ is not small. Quantitatively, $|f(t)|$ is at least a constant times the distance from t to the nearest root, and the constant depends on how close the roots are to each other, which the discriminant controls. Silverman's last part asks for a constant that depends only on the degree of f and on the projective height of its vector of coefficients. No such constant exists: dividing f by an integer N leaves this height unchanged, but at an archimedean place it divides $|f(t)|_v$ by N. So Problem 1.6(5) asks for a constant that depends on the affine height of the coefficients instead.

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Definition 1.1 (Discriminant)**.** Let L be a field and $f = a_0T^n + a_1T^{n-1} + \cdots + a_n \in L[T]$ with $n \geqslant 1$ and $a_0 \neq 0$. Over an algebraic closure of L, $f = a_0 \prod_{i=1}^n (T - \xi_i)$. The discriminant of f is

$$D(f) \coloneqq a_0^{2n-2} \prod_{i<j} (\xi_i - \xi_j)^2.$$

It is a polynomial with integer coefficients in $a_0, \ldots, a_n$, by the theorem on symmetric polynomials, since each ξ_i occurs in $\prod_{i<j}(\xi_i - \xi_j)^2$ to degree at most $2n-2$. So $D(f) \in L$, and $D(f) \in \mathbb{Z}$ when $f \in \mathbb{Z}[T]$. It is nonzero if and only if the roots of f are distinct.

Definition 1.2 (Mahler measure [HS00, §B.7])**.** The Mahler measure of $f = a_0 \prod_{i=1}^n (T - \xi_i) \in \mathbb{C}[T]$, with $a_0 \neq 0$, is

$$M(f) \coloneqq |a_0| \prod_{i=1}^n \max(1, |\xi_i|).$$

Lemma 1.3 (Mahler measure and coefficients [HS00, Lemma B.7.3.1 and the formula after it])**.** *Let $f = a_0T^n + \cdots + a_n \in \mathbb{C}[T]$ with $a_0 \neq 0$. Then $M(f) \leqslant (n+1)^{1/2} \max_k |a_k|$.*

Proposition 1.4 (Gauss's lemma [HS00, proof of Proposition B.7.3])**.** *Let v be a non-archimedean place of K. For a polynomial g with coefficients in $\overline{K}$, let $|g|_v$ be the maximum of the absolute values of its coefficients. Then $|gh|_v = |g|_v |h|_v$. In particular, if $f = a_0 \prod_{i=1}^n (T - \xi_i)$, then*

$$|f|_v = |a_0|_v \prod_{i=1}^n \max(1, |\xi_i|_v).$$

Proposition 1.5 (Hadamard's inequality)**.** *For $z = (z_1, \ldots, z_m) \in \mathbb{C}^m$, let $\|z\| \coloneqq \left(\sum_k |z_k|^2\right)^{1/2}$. Let V be an $m \times m$ complex matrix with rows $r_1, \ldots, r_m$. Then $|\det V| \leqslant \prod_{i=1}^m \|r_i\|$.*

Problem 1.6 (Polynomials are small only near their roots)**.**
Let $f = a_0T^n + a_1T^{n-1} + \cdots + a_n \in K[T]$ be a polynomial of degree $n \geqslant 2$, with $a_0 a_n \neq 0$ and with distinct roots $\xi_1, \ldots, \xi_n \in \overline{K}$. So $f = a_0 \prod_{i=1}^n (T - \xi_i)$, and $D(f) \neq 0$. (Silverman allows $n = 1$, where $f(t) = a_0(t - \xi_1)$ and every statement below is immediate.)

(1) Let $v \in \mathcal{P}_K$, and let $\varepsilon_v \coloneqq 1/2$ if v is archimedean and $\varepsilon_v \coloneqq 1$ if v is non-archimedean. Let $t \in \overline{K}$, and let ξ_i be a root of f nearest to t, that is, $|t - \xi_i|_v = \min_j |t - \xi_j|_v$. Show that

$$|f(t)|_v \geqslant \varepsilon_v^{n-1} |f'(\xi_i)|_v |t - \xi_i|_v.$$

Solution. **For $j \neq i$, $\varepsilon_v |\xi_i - \xi_j|_v \leqslant |t - \xi_j|_v$.** For $j \neq i$, the choice of ξ_i gives

$$|\xi_i - \xi_j|_v \leqslant |\xi_i - t|_v + |t - \xi_j|_v \leqslant 2|t - \xi_j|_v,$$

and if v is non-archimedean, then even $|\xi_i - \xi_j|_v \leqslant \max(|\xi_i - t|_v, |t - \xi_j|_v) = |t - \xi_j|_v$. So $\varepsilon_v |\xi_i - \xi_j|_v \leqslant |t - \xi_j|_v$ in both cases. By the product rule, $f'(\xi_i) = a_0 \prod_{j \neq i} (\xi_i - \xi_j)$. So

$$|f(t)|_v = |a_0|_v |t - \xi_i|_v \prod_{j \neq i} |t - \xi_j|_v \geqslant \varepsilon_v^{n-1} |a_0|_v |t - \xi_i|_v \prod_{j \neq i} |\xi_i - \xi_j|_v = \varepsilon_v^{n-1} |f'(\xi_i)|_v |t - \xi_i|_v.$$

(2) In this item and the next, $K = \mathbb{Q}$, $f \in \mathbb{Z}[T]$, and $|\ |$ is the usual absolute value on $\mathbb{C}$. Show that every root ξ_i of f satisfies

$$|f'(\xi_i)| \geqslant (n-1)^{-(n-1)/2} M(f)^{-(n-2)}.$$

Solution. **Combine $|D(f)| \geqslant 1$ with Hadamard's inequality.** Number the roots so that $i = 1$, and let V be the Vandermonde matrix of $\xi_2, \dots, \xi_n$, the $(n-1) \times (n-1)$ matrix

$$V \coloneqq \begin{pmatrix} 1 & \xi_2 & \xi_2^2 & \cdots & \xi_2^{n-2} \\ 1 & \xi_3 & \xi_3^2 & \cdots & \xi_3^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \xi_n & \xi_n^2 & \cdots & \xi_n^{n-2} \end{pmatrix}.$$

For $n = 2$, $V = (1)$. As $f \in \mathbb{Z}[T]$ has distinct roots, $D(f)$ is a nonzero integer. So

$$1 \leqslant |D(f)|^{1/2}$$

Split the pairs $i < j$ in the definition of $D(f)$ according to whether $i = 1$:

$$= |a_0| \prod_{j=2}^{n} |\xi_1 - \xi_j| \cdot |a_0|^{n-2} \prod_{2 \leqslant i < j \leqslant n} |\xi_i - \xi_j|$$

As $f'(\xi_1) = a_0 \prod_{j=2}^{n} (\xi_1 - \xi_j)$ and $|\det V| = \prod_{2 \leqslant i < j \leqslant n} |\xi_i - \xi_j|$,

$$= |f'(\xi_1)| |a_0|^{n-2} |\det V|$$

By Proposition 1.5, $|\det V|$ is at most the product of the norms of the rows of V:

$$\leqslant |f'(\xi_1)| |a_0|^{n-2} \prod_{i=2}^{n} \Big(\sum_{k=0}^{n-2} |\xi_i|^{2k} \Big)^{1/2}$$

Each of the $n-1$ terms of the sum is at most $\max(1, |\xi_i|)^{2(n-2)}$:

$$\leqslant (n-1)^{(n-1)/2} |f'(\xi_1)| \Big(|a_0| \prod_{i=2}^{n} \max(1, |\xi_i|) \Big)^{n-2}$$

Finally, $\max(1, |\xi_1|) \geqslant 1$ and $n - 2 \geqslant 0$:

$$\leqslant (n-1)^{(n-1)/2} |f'(\xi_1)| M(f)^{n-2}.$$

Dividing by $(n-1)^{(n-1)/2} M(f)^{n-2}$ gives the bound.

(3) Let $A \coloneqq \max(|a_0|, \dots, |a_n|)$. Show that every rational number t satisfies

$$|f(t)| \geqslant \frac{\min\big(|t - \xi_1|, \dots, |t - \xi_n|\big)}{2^{n-1} (n+1)^{n-3/2} A^{n-2}}.$$

Solution. **Combine (1) and (2) with $M(f) \leqslant (n+1)^{1/2} A$.** Let ξ_i be a root nearest to t. By (1), with $\varepsilon_\infty = 1/2$, (2) and Lemma 1.3,

$$|f(t)| \geqslant \frac{|f'(\xi_i)| |t - \xi_i|}{2^{n-1}} \geqslant \frac{|t - \xi_i|}{2^{n-1} (n-1)^{(n-1)/2} (n+1)^{(n-2)/2} A^{n-2}} \geqslant \frac{|t - \xi_i|}{2^{n-1} (n+1)^{n-3/2} A^{n-2}}.$$

Remark. Silverman states the weaker bound $|f(t)| \geqslant (2n^2 A)^{-n} \min_i |t - \xi_i|$. It follows, since $A \geqslant 1$ as a_0 is a nonzero integer, and $n + 1 \leqslant n^2$ as $n \geqslant 2$, so

$$2^{n-1} (n+1)^{n-3/2} A^{n-2} \leqslant 2^n (n^2)^n A^n = (2n^2 A)^n.$$

The proof never uses $t \in \mathbb{Q}$, and the argument of (1) works verbatim for every $t \in \mathbb{C}$. Rationality enters only in Problem 2.5, through Roth's theorem.

(4) Let $S \subseteq \mathcal{P}_K$ be a finite set of places. Show that there is a constant $C_f > 0$, depending only on f, such that every $t \in K$ satisfies

$$\prod_{v \in S} \min\bigl(1, |f(t)|_v^{n_v}\bigr) \geqslant C_f \prod_{v \in S} \min_{1 \leqslant i \leqslant n} \min\bigl(1, |t - \xi_i|_v^{n_v}\bigr).$$

SOLUTION. **One can take $C_f = \prod_{v \in S} \min(1, c_v)^{n_v}$ with $c_v = \varepsilon_v^{n-1} \min_i |f'(\xi_i)|_v$.** For $v \in S$, let $\delta_v := \min_i |t - \xi_i|_v$. As the roots are distinct, $f'(\xi_i) \neq 0$ for every i, so $c_v > 0$. By (1), $|f(t)|_v \geqslant c_v \delta_v$. For real numbers $x, y \geqslant 0$, $\min(1, xy) \geqslant \min(1, x) \min(1, y)$. So

$$\min\bigl(1, |f(t)|_v\bigr) \geqslant \min(1, c_v) \min(1, \delta_v).$$

Raising to the power n_v and multiplying over $v \in S$ gives the inequality, since $\min(1, \delta_v)^{n_v} = \min_i \min(1, |t - \xi_i|_v^{n_v})$.

(5) Let $H := H_K([1 : a_0 : \cdots : a_n])$. Find a constant C_f for (4) that depends only on n, $[K : \mathbb{Q}]$ and H.

SOLUTION. **One can take $C_f = (2n+2)^{-2n[K:\mathbb{Q}]} H^{-(2n-3)}$.** For $v \in \mathcal{P}_K$, let $|f|_v := \max_k |a_k|_v$, let $M_v(f) := |a_0|_v \prod_i \max(1, |\xi_i|_v)$, and let $e_v := 1$ if v is archimedean and $e_v := 0$ otherwise. If v is archimedean, $|\ |_v$ on $\overline{K}$ comes from an embedding of $\overline{K}$ in $\mathbb{C}$, so $M_v(f) \leqslant (n+1)^{1/2} |f|_v$ by Lemma 1.3. If v is non-archimedean, $M_v(f) = |f|_v$ by Proposition 1.4.

The derivative. Number the roots so that $i = 1$. The two equalities in the solution of (2) hold for every absolute value:

$$|D(f)|_v^{1/2} = |f'(\xi_1)|_v |a_0|_v^{n-2} \prod_{2 \leqslant i < j \leqslant n} |\xi_i - \xi_j|_v.$$

At an archimedean v, the rest of that chain bounds $|a_0|_v^{n-2} \prod_{2 \leqslant i < j \leqslant n} |\xi_i - \xi_j|_v$ by $(n-1)^{(n-1)/2} M_v(f)^{n-2}$. At a non-archimedean v, $|\xi_i - \xi_j|_v \leqslant \max(1, |\xi_i|_v) \max(1, |\xi_j|_v)$, and each $i \geqslant 2$ occurs in $n-2$ pairs, so it is bounded by $M_v(f)^{n-2}$. So in both cases

$$|D(f)|_v^{1/2} \leqslant (n-1)^{e_v(n-1)/2} |f'(\xi_1)|_v M_v(f)^{n-2} \leqslant \bigl((n-1)^{(n-1)/2} (n+1)^{(n-2)/2}\bigr)^{e_v} |f'(\xi_1)|_v |f|_v^{n-2}.$$

Let $\delta_v := \min_i |t - \xi_i|_v$ and $\gamma_v := \bigl(2^{n-1}(n+1)^{n-3/2}\bigr)^{e_v}$. With (1), applied to a root ξ_i nearest to t, this gives

$$|f(t)|_v \geqslant \frac{\delta_v |D(f)|_v^{1/2}}{\gamma_v |f|_v^{n-2}}.$$

As $\gamma_v \geqslant 1$,

$$\min\bigl(1, |f(t)|_v\bigr) \geqslant \gamma_v^{-1} \min(1, \delta_v) \min\bigl(1, |D(f)|_v\bigr)^{1/2} \max\bigl(1, |f|_v\bigr)^{-(n-2)}.$$

The product over S. Raise this to the power n_v and multiply over $v \in S$. Three factors need a bound.

- As $\sum_{v \in \mathcal{P}_K^\infty} n_v = [K : \mathbb{Q}]$, $\prod_{v \in S} \gamma_v^{n_v} \leqslant \bigl(2^{n-1}(n+1)^{n-3/2}\bigr)^{[K:\mathbb{Q}]}$.
- The number $\max(1, |f|_v)^{n_v}$ is the factor of H at v, and every factor of H is at least 1. So $\prod_{v \in S} \max(1, |f|_v)^{n_v} \leqslant H$.
- Let $D := D(f) \in K^\times$. Then $\min(1, |D|_v) = \max(1, |D^{-1}|_v)^{-1}$, so as for H,

$$\prod_{v \in S} \min\bigl(1, |D|_v\bigr)^{n_v} \geqslant H_K(D^{-1})^{-1} = H_K(D)^{-1},$$

because $[D^{-1} : 1] = [1 : D]$. Hadamard's inequality for the full Vandermonde matrix of $\xi_1, \dots, \xi_n$ at the archimedean places, and the ultrametric inequality at the others, give $|D|_v \leqslant n^{ne_v} M_v(f)^{2n-2} \leqslant \big(n^n(n+1)^{n-1}\big)^{e_v} |f|_v^{2n-2}$. Hence $H_K(D) \leqslant \big(n^n(n+1)^{n-1}\big)^{[K:\mathbb{Q}]} H^{2n-2}$. Altogether, $\prod_{v\in S} \min(1, |f(t)|_v^{n_v})$ is at least $\prod_{v\in S} \min(1, \delta_v^{n_v})$ times

$$\begin{aligned}&\big(2^{n-1}(n+1)^{n-3/2}\big)^{-[K:\mathbb{Q}]} \big(n^n(n+1)^{n-1}\big)^{-[K:\mathbb{Q}]/2} H^{-(n-1)} H^{-(n-2)} \\ &\quad = \big(2^{n-1} n^{n/2} (n+1)^{(3n-4)/2}\big)^{-[K:\mathbb{Q}]} H^{-(2n-3)}.\end{aligned}$$

Finally $2^{n-1} n^{n/2} (n+1)^{(3n-4)/2} \leqslant 2^{2n}(n+1)^{2n} = (2n+2)^{2n}$.

Remark. Silverman asks for a constant depending only on n and the projective height $H_K([a_0 : \cdots : a_n])$. There is none. Over $K = \mathbb{Q}$, with $S = \{\infty\}$, let $f = (T^2 - 2)/N$ for an integer $N \geqslant 2$ and $t = 0$. The left side is $\min(1, 2/N) = 2/N$, and the right side is $C_f \min(1, \sqrt{2}) = C_f$, so $C_f \leqslant 2/N$. But $H_{\mathbb{Q}}([1/N : 0 : -2/N]) = H_{\mathbb{Q}}([1 : 0 : -2]) = 2$ for every N, while the affine height $H_{\mathbb{Q}}([1 : 1/N : 0 : -2/N]) = N$ grows.

2. Thue's equation

Problem 2.5 is [Sil09, Exercise 9.6(a)–(b)], with Roth's theorem in the form of [HS00, Theorem D.2.1]. If y is large, then at each place of S the ratio x/y of a solution is close to a root of $F(T, 1)$, and this root may change from place to place. Theorem 2.3 has a single algebraic number ξ, but it holds for every choice of the embeddings ι_v, so ξ can stand for a different conjugate at each place. So Problem 2.5(3) assumes that F has an irreducible factor of degree at least 3, whose roots are conjugate. Silverman states that part for every F of degree at least 3 with nonzero discriminant.

Definition 2.1 (Binary forms)**.** A binary form of degree d over K is a nonzero homogeneous polynomial $F \in K[X, Y]$ of degree d. Over $\overline{K}$ it is a product of linear forms,

$$F = \prod_{i=1}^{d} L_i, \qquad L_i = \beta_i X - \alpha_i Y.$$

Let $\Delta_{ij} \coloneqq \alpha_i\beta_j - \alpha_j\beta_i$. The discriminant of F is $\prod_{i<j} \Delta_{ij}^2$. It does not depend on the factorization, and if $F(1, 0) \neq 0$, it is the discriminant $D(f)$ of $f(T) \coloneqq F(T, 1)$. It is nonzero if and only if no two of the L_i are proportional.

Definition 2.2 (S-integers [Sil09, proof of Proposition VIII.1.6])**.** Let $S \subseteq \mathcal{P}_K$ be a finite set containing $\mathcal{P}_K^\infty$. The ring of S-integers of K is

$$\mathcal{O}_S \coloneqq \{a \in K : |a|_v \leqslant 1 \text{ for all } v \notin S\}.$$

Theorem 2.3 (Roth [HS00, Theorem D.2.1])**.** *Let $S \subseteq \mathcal{P}_K$ be a finite set, $\xi \in \overline{K}$ and $\varepsilon > 0$. Then only finitely many $t \in K$ satisfy*

$$\prod_{v\in S} \min\big(1, |t - \xi|_v^{n_v}\big) \leqslant H_K(t)^{-(2+\varepsilon)}.$$

Theorem 2.4 (Northcott [Sil09, Theorem VIII.5.11])**.** *For every $B > 0$, only finitely many $t \in K$ satisfy $H_K(t) \leqslant B$.*

Problem 2.5 (Thue's equation)**.**
Let $F \in K[X,Y]$ be a binary form of degree $d \geqslant 3$ with nonzero discriminant, and let $b \in K^\times$.
(1) Let $K = \mathbb{Q}$, and suppose that $F \in \mathbb{Z}[X,Y]$ and $b \in \mathbb{Z}$. Show that $F(x,y) = b$ has only finitely many solutions $(x,y) \in \mathbb{Z}^2$.

Solution. **For large y, x/y is too close to a root of $F(T,1)$, as $d \geqslant 3$.** The proof has four steps.

Step 1. Changes of variables make $F(1,0)F(0,1) \neq 0$.
Step 2. For a solution with $|y|$ large, $t := x/y$ is within $C'H_{\mathbb{Q}}(t)^{-d}$ of a root of $F(T,1)$, for a constant C'.
Step 3. Roth's theorem leaves only finitely many such t.
Step 4. Each t comes from finitely many solutions, and so do the small values of y.

Step 1: reduction. For $m \in \mathbb{Z}$, the maps $(x,y) \mapsto (x, y+mx)$ and $(x,y) \mapsto (x+my, y)$ are bijections of $\mathbb{Z}^2$, and composing F with them gives binary forms of degree d with integer coefficients and nonzero discriminant, since an invertible linear substitution keeps non-proportional linear forms non-proportional. As $F(X,Y) = X^dF(1,Y/X)$, the polynomial $F(1,T)$ is nonzero, so $F(1,m) \neq 0$ for some $m \in \mathbb{Z}$. Then $G(X,Y) := F(X,Y+mX)$ has $G(1,0) \neq 0$. Likewise $G(k,1) \neq 0$ for some $k \in \mathbb{Z}$, and $G(X+kY,Y)$ takes the values $G(1,0)$ and $G(k,1)$ at $(1,0)$ and $(0,1)$. So we may assume that $a_0 := F(1,0)$ and $a_d := F(0,1)$ are nonzero. Then $f(T) := F(T,1) = a_0T^d + \cdots + a_d$ has degree d, $a_0a_d \neq 0$, and $D(f) \neq 0$ by Definition 2.1. Let $\xi_1, \dots, \xi_d \in \overline{\mathbb{Q}}$ be its roots.
Step 2: close to a root. Let $A := \max_k |a_k|$, $\gamma := 2^{d-1}(d+1)^{d-3/2}A^{d-2}$, $C := \gamma|b|$ and $R := \max_i|\xi_i|$. Let (x,y) be a solution with $|y|^d \geqslant C$, and $t := x/y$. Then $f(t) = y^{-d}F(x,y) = b/y^d$, so by Problem 1.6(3) some root ξ_i satisfies

$$|t - \xi_i| \leqslant \gamma|f(t)| = C|y|^{-d} \leqslant 1.$$

So $|x| \leqslant (R+1)|y|$, and $H_{\mathbb{Q}}(t) \leqslant \max(|x|,|y|) \leqslant (R+1)|y|$, since x and y need not be coprime. Let $C' := C(R+1)^d$. Then

$$|t - \xi_i| \leqslant C'H_{\mathbb{Q}}(t)^{-d}.$$

Step 3: finitely many t. As $d \geqslant 3$ and $H_{\mathbb{Q}}(t) \geqslant 1$, Step 2 gives

$$|t - \xi_i| \leqslant C'H_{\mathbb{Q}}(t)^{-1/2} \cdot H_{\mathbb{Q}}(t)^{-5/2}.$$

If $H_{\mathbb{Q}}(t) \geqslant C'^2$, then $\min(1, |t-\xi_i|) \leqslant H_{\mathbb{Q}}(t)^{-5/2}$, and Theorem 2.3, with $K = \mathbb{Q}$, $S = \{\infty\}$ and $\varepsilon = 1/2$, applied to each of the d roots, leaves only finitely many such t. Only finitely many $t \in \mathbb{Q}$ have $H_{\mathbb{Q}}(t) < C'^2$, as their numerators and denominators are bounded. So the solutions with $|y|^d \geqslant C$ give only finitely many values of t.
Step 4: back to the solutions. For each value of t from Step 3, $f(t) = b/y^d \neq 0$, so $|y|^d = |b/f(t)|$ leaves at most two values of y, and $x = ty$. The solutions with $|y|^d < C$ have finitely many values of y. For each, the polynomial $F(X,y) - b$ has leading coefficient $a_0 \neq 0$, so only finitely many x satisfy $F(x,y) = b$.
Remark. The hypothesis $d \geqslant 3$ is needed: the equation $x^2 - 2y^2 = 1$ has the infinitely many solutions given by $x + y\sqrt{2} = \pm(3+2\sqrt{2})^k$, $k \in \mathbb{Z}$. For $d = 2$ the argument gives $|t - \xi_i| \leqslant C'H_{\mathbb{Q}}(t)^{-2}$, which Dirichlet's theorem allows infinitely often [Sil09, Proposition IX.1.2].

(2) `Magma` solves Thue equations. The code below lists all the integer solutions of $x^3 - 2y^3 = b$ for $1 \leqslant b \leqslant 6$. Explain why the argument of (1) could not produce such a list.

```
R<x> := PolynomialRing(Integers());
```

```
t := Thue(x^3 - 2);                    // the form X^3 - 2*Y^3
[ <b, Solutions(t, b)> : b in [1..6] ];
// [ <1, [[-1, -1], [1, 0]]>, <2, [[0, -1]]>, <3, [[-5, -4], [1, -1]]>,
//   <4, []>, <5, []>, <6, [[2, 1]]> ]
```

Solution. **Roth's theorem gives no bound for the height of the exceptions.** The argument of (1) bounds $|y|$ by an explicit constant except for the solutions with x/y in the finite set of Theorem 2.3. That theorem says the set is finite, but its proof gives no bound for the heights of its elements [Sil09, Remark IX.1.6]. So a search can find solutions, but never knows when it has found them all.

Remark. `Magma` uses instead Baker's lower bounds for linear forms in logarithms, which give an explicit, if huge, bound for the solutions, and then reduces it until a search is feasible [Sil09, §IX.5].

(3) Let $S \subseteq \mathcal{P}_K$ be a finite set containing $\mathcal{P}_K^\infty$, and suppose that F has an irreducible factor $G \in K[X,Y]$ of degree $g \geqslant 3$. Show that $F(x,y) = b$ has only finitely many solutions $(x,y) \in \mathcal{O}_S \times \mathcal{O}_S$.

Solution. **Conjugating the nearest roots to one root reduces to Roth's theorem.** *Reductions.* The changes of variables of (1) are also bijections of $\mathcal{O}_S \times \mathcal{O}_S$. As there, $F(1,m) \neq 0$ for some $m \in \mathbb{Z}$, so after replacing F by $F(X, Y+mX)$ and G by $G(X, Y+mX)$ we may assume that $a_0 \coloneqq F(1,0)$ is nonzero. Then $G(1,0) \neq 0$, as G divides F. Write $F = GG'$ with $G' \in K[X,Y]$. A nonzero element of K has $|\ |_v > 1$ at only finitely many places, and enlarging S only enlarges $\mathcal{O}_S$, so we may assume that the coefficients of G' lie in $\mathcal{O}_S$. Since $G(1,0) \neq 0$, the polynomial $G(T,1)$ has degree g, and it is irreducible over K, as a factorization of it would homogenize to one of G. Its roots $\xi_1, \dots, \xi_g \in \overline{K}$ are distinct, since F has nonzero discriminant. They are nonzero, since $G(0,1) \neq 0$: otherwise X would divide G, which is irreducible of degree at least 3. If $y = 0$, then $a_0 x^d = b$ leaves at most d values of x. So let (x,y) be a solution with $y \neq 0$, and let $t \coloneqq x/y$ and $M_v \coloneqq \max(|x|_v, |y|_v)$ for $v \in \mathcal{P}_K$.
The places of S. The element $c \coloneqq G'(x,y)$ lies in $\mathcal{O}_S$, and $c \neq 0$ because $G(x,y)c = b$. By the product formula, $\prod_{v\in S} |c|_v^{n_v} = \prod_{v\notin S} |c|_v^{-n_v} \geqslant 1$, so

$$\prod_{v\in S} |G(x,y)|_v^{n_v} \leqslant B \coloneqq \prod_{v\in S} |b|_v^{n_v}.$$

A local bound. Fix $v \in S$, and let $a \coloneqq G(1,0)$, $R_v \coloneqq \max(1, 2\max_i |\xi_i|_v)$, $m_v \coloneqq \min_i |\partial_T G(\xi_i, 1)|_v$, which is positive as the roots are distinct, and $\delta_v \coloneqq \min_i \min(1, |t-\xi_i|_v)$. Let $\kappa_v \coloneqq \min(2^{-g}|a|_v, \varepsilon_v^{g-1} R_v^{-g} m_v)$, with ε_v as in Problem 1.6(1). It is positive and does not depend on the solution. We show that $|G(x,y)|_v \geqslant \kappa_v M_v^g \delta_v$. Here $G(x,y) = a\prod_i (x - \xi_i y)$. If $|x|_v > R_v |y|_v$, then $M_v = |x|_v$ and $|x - \xi_i y|_v \geqslant |x|_v - |\xi_i|_v |y|_v \geqslant |x|_v/2$ for every i, so $|G(x,y)|_v \geqslant 2^{-g}|a|_v M_v^g$, which is enough as $\delta_v \leqslant 1$. Otherwise $M_v \leqslant R_v |y|_v$, and Problem 1.6(1), applied to $G(T,1)$, gives

$$|G(x,y)|_v = |y|_v^g |G(t,1)|_v \geqslant R_v^{-g} M_v^g \cdot \varepsilon_v^{g-1} m_v \delta_v.$$

The height. For $v \notin S$, $M_v \leqslant 1$, so $H_K(t) = H_K([x:y]) \leqslant \prod_{v\in S} M_v^{n_v}$. Raising the local bounds to the powers n_v and multiplying, with $\kappa \coloneqq \prod_{v\in S} \kappa_v^{n_v}$,

$$\prod_{v\in S} \min_i \min\bigl(1, |t-\xi_i|_v^{n_v}\bigr) = \prod_{v\in S} \delta_v^{n_v} \leqslant \frac{B}{\kappa} \prod_{v\in S} M_v^{-gn_v} \leqslant \frac{B}{\kappa} H_K(t)^{-g}.$$

One root. For each $v \in S$, choose $i(v)$ with $|t - \xi_{i(v)}|_v = \min_i |t-\xi_i|_v$. There are only finitely many maps $i\colon S \to \{1, \dots, g\}$, so it suffices to fix one and show that only finitely many t have

it. As $G(T,1)$ is irreducible over K, for each $v \in S$ there is $\sigma_v \in \mathrm{Aut}(\overline{K}/K)$ with $\sigma_v(\xi_1) = \xi_{i(v)}$. Replace each ι_v by the K-embedding $\iota_v \circ \sigma_v$, and write $|\ |'_v$ for the resulting extension of $|\ |_v$ to $\overline{K}$, so that $|z|'_v = |\sigma_v(z)|_v$. As $\sigma_v(t) = t$, $|t - \xi_1|'_v = |t - \xi_{i(v)}|_v$. With $C \coloneqq B/\kappa$, and since $g \geqslant 3$ and $H_K(t) \geqslant 1$,

$$\prod_{v \in S} \min\bigl(1, (|t-\xi_1|'_v)^{n_v}\bigr) \leqslant C H_K(t)^{-g} \leqslant C H_K(t)^{-1/2} \cdot H_K(t)^{-5/2}.$$

If $H_K(t) \geqslant C^2$, the right side is at most $H_K(t)^{-5/2}$, and Theorem 2.3, with the embeddings $\iota_v \circ \sigma_v$, $\xi = \xi_1$ and $\varepsilon = 1/2$, leaves only finitely many t. By Theorem 2.4, only finitely many t have $H_K(t) < C^2$.

Back to the solutions. For each value of t, $f(t) \coloneqq F(t,1) = b/y^d \neq 0$, so $y^d = b/f(t)$ leaves at most d values of y, and $x = ty$.

Remark. The irreducible factor G is used only to move the nearest roots at the places of S to a single root. If S is a single place, for instance if K is $\mathbb{Q}$ or an imaginary quadratic field and $S = \mathcal{P}_K^\infty$, there is nothing to move: the same argument, with $G = F$ and the nearest root fixed by the pigeonhole principle, works for every F of degree at least 3 with nonzero discriminant, as in (1). With several places, the general case needs Roth's theorem with a different algebraic number at each place, which Theorem 2.3 does not give. For $F = XY(X+Y)$, for instance, x/y may be close to 0 at some places of S and to -1 at others, and these roots are not conjugate. Silverman's statement still holds for such F, but its proof needs a further idea.

References

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